CS & IT ENGINEERING



Computer Networks

Error Control Methods

DPP 01 Discussion Notes



[MCQ]

#Q. A building running CSMA-CD protocol has a bandwidth of 512 Mbps and a distance of 2 kilometres, and then the minimum data size is determined in order to detect a collision. Assume that the signal speed is 2,00,000 km/s

A 1000 bytes

B 1250 bytes

C 1280 bytes

D 1024 bytes

CSMA/CD collision to detect collision it required $\leq 2PT$

$T_{fr} \geq 2PT$

$PT = \frac{l}{V} = \frac{2}{2 \times 10^5} = 10^{-5}$ sec.

$\frac{\text{Datasize}}{\text{Bandwidth}} \geq 2 \times 10^{-5}$

$\text{DataSize} \geq 2 \times 512 \times 10^6 \times 10^{-5}$

Data size $\geq$ 10240 bits.

Data size $\geq$ 1280 bytes.

[MCQ]

#Q. A transmission channel operates at 10 kbps bandwidth. Distance between two Stations in the channel are 2 km. Assume that the speed of propagation of a bit is 200 m/sec. What is the frame length if stop-and-wait ARQ is used with 50% efficiency?

A 25 KB

B 50 KB

C 100 KB

D 200 KB

$$\eta = \frac{T_d(\text{frame})}{T_d(\text{frame}) + 2P_d + Q_d + P_{proc} + T_d(\text{ack})}$$

$Q_d = 0$, $P_{proc} = 0$, $T_d(\text{ack}) = 0$

$$\eta = \frac{T_d(\text{frame})}{T_d(\text{frame}) + 2P_d}$$

$$= \frac{1}{1 + 2 \cdot \frac{P_d}{T_d(\text{frame})}}$$

$$.5 = \frac{1}{1 + 2 \cdot \frac{10}{\frac{D}{10Kb}}} =$$

$$P_d = \frac{l}{v} = \frac{2000\text{ m}}{200\text{ m/sec}} = 10\text{ sec}$$

$$T_d = \frac{\text{Data size}(D)}{\text{Bandwidth}} = \frac{D}{10\text{ Kbps}}$$

$$= \frac{1}{1 + \frac{2 \times 100Kb}{D}} = \frac{D}{D + 200Kb}$$

$$.5D + 100\text{ Kb} = D$$

$$.5D = 100\text{ Kb}$$

$$D = 200\text{ Kbits}$$

$$= 25\text{ KB}.$$

[MCQ]

#Q. Suppose a 10 Mbps microwave link exists between a geostationary satellite and its base station on Earth. The satellite takes a digital photo every minute and sends it to the base station. Assume a propagation speed of 2.4×10^8 meters/sec. Let x denote the size of the photo. What is the minimum value of x for the microwave link to be continuously transmitting? Assume a geostationary satellite is 36,000 kilometres away from the earth's surface.

A. 6 Mbits

B. 4 Mbits

C. 3 Mbits

D. 5 Mbits

$T_t = 2 P_d$

$P_d = \frac{l}{v} = \frac{36000 \times 10^3}{2.4 \times 10^8} = \frac{36}{2.4 \times 10^2}$ sec

$\frac{\text{Datasize}}{\text{Bandwidth}} = 2 \times \frac{36}{2.4 \times 10^2}$

$\text{Datasize} = \frac{2 \times 36 \times 10 \times 10^6}{2.4 \times 10^2} = 3$ Mbits

[MCQ]

#Q. Consider a 100 Mbps ethernet with a maximum distance of 400 meters. What is the smallest packet length for which you can achieve an efficiency of 80%? Assume velocity is equal to the speed of light.

A 2022 bits

B 1066 bits

C 1520 bits

D None

$$\eta = \frac{T_d(\text{frame})}{T_d(\text{frame}) + 2P_d + Q_d + P_{rd} + T_d(\text{ack})}$$

($Q_d \downarrow 0$, $P_{rd} \downarrow 0$, $T_d(\text{ack}) \downarrow 0$)

$$= \frac{T_d(\text{frame})}{T_d(\text{frame}) + 2P_d} = \frac{1}{1 + 2 \cdot \frac{P_d}{T_d(\text{frame})}}$$

$$P_d = \frac{l}{v} = \frac{400'}{3\times10^8 \text{ m/s}} = \frac{4}{3} \times 10^{-6} \text{ sec}$$

$$T_d(\text{frame}) = \frac{\text{Data packed Size}}{\text{Bandwidth}} = \frac{\text{Data Size}}{100\text{Mb}}$$

$$\eta = \frac{1}{1 + 2 \cdot \frac{P_d}{T_d(\text{frame})}}$$

$$\cdot 8 = \frac{1}{1 + 2 \cdot \frac{\frac{1}{3} \times 10^{-6}}{\frac{\text{Data}}{100\text{Mb}}}} = \frac{1}{1 + \frac{8}{3} \times \frac{100}{\text{Data}}} = \frac{3\,\text{Data}}{3\,\text{Data} + 800}$$

$$2.4\,\text{Data} + 640 = 3 \times \text{Data}$$

$$\text{Data} = \frac{640}{\cdot 6}$$

$$= \frac{3200}{3} \simeq 1066 \text{ bits}$$

[MCQ]

#Q. If the transmission time is 1 ms and the propagation time is 1.5 ms, what is the efficiency of stop and wait protocol?

A. 0.25

B. 0.05

C. 0.30

D. 0.50

$$\eta = \frac{1}{1 + 2 \cdot \frac{P_d}{T_d}}$$

$$= \frac{1}{1 + 2 \times \frac{1.5}{1}} = \frac{1}{4} = 0.25$$

[MCQ]

#Q. In the Go-Back-N protocol, if the maximum window size is 127, what is the range of the sequence number?

A 0 to 127

B 0 to 128

C 1 to 127

D 1 to 128

Go back N maximum sender window size + Receiver window size = total sequence no required

127 + 1 = 128

↓ Range (0 – 127)

[NAT]

#Q. Station 'A' uses 64 Byte packets to transmit messages to station 'B' using a sliding window protocol. The round trip delay between A and B is 80 milliseconds, and the bottleneck bandwidth on the path between 'A' and 'B' is 128 Kbps. What is the sender window size for maximum efficiency?

$2P_d = 80$, $B = 128$ Kbps.

$P_d = 40$ msec.

$B = \frac{}{}$

$T_d =$

Sender window size for maximum efficiency

$= 1 + 2a$ $\left[a = \frac{P_d}{T_d}\right]$

$= 1 + 2 \cdot \frac{40}{4} = 21$ Ans

128 kbit $\longrightarrow$ 1 sec

1 bit $\longrightarrow$ $\frac{1}{128 \times 10^3} = \frac{1}{128}$ msec.

64 x 8 bit $\longrightarrow$ $\frac{1}{128} \times 64 \times 8 = 4$ msec.

[MCQ]

#Q. Consider the following message M=1010001011. The cyclic redundancy check(CRC) for this message using the divisor polynomial $x^5 + x^3 + x^2 + 1$ is:

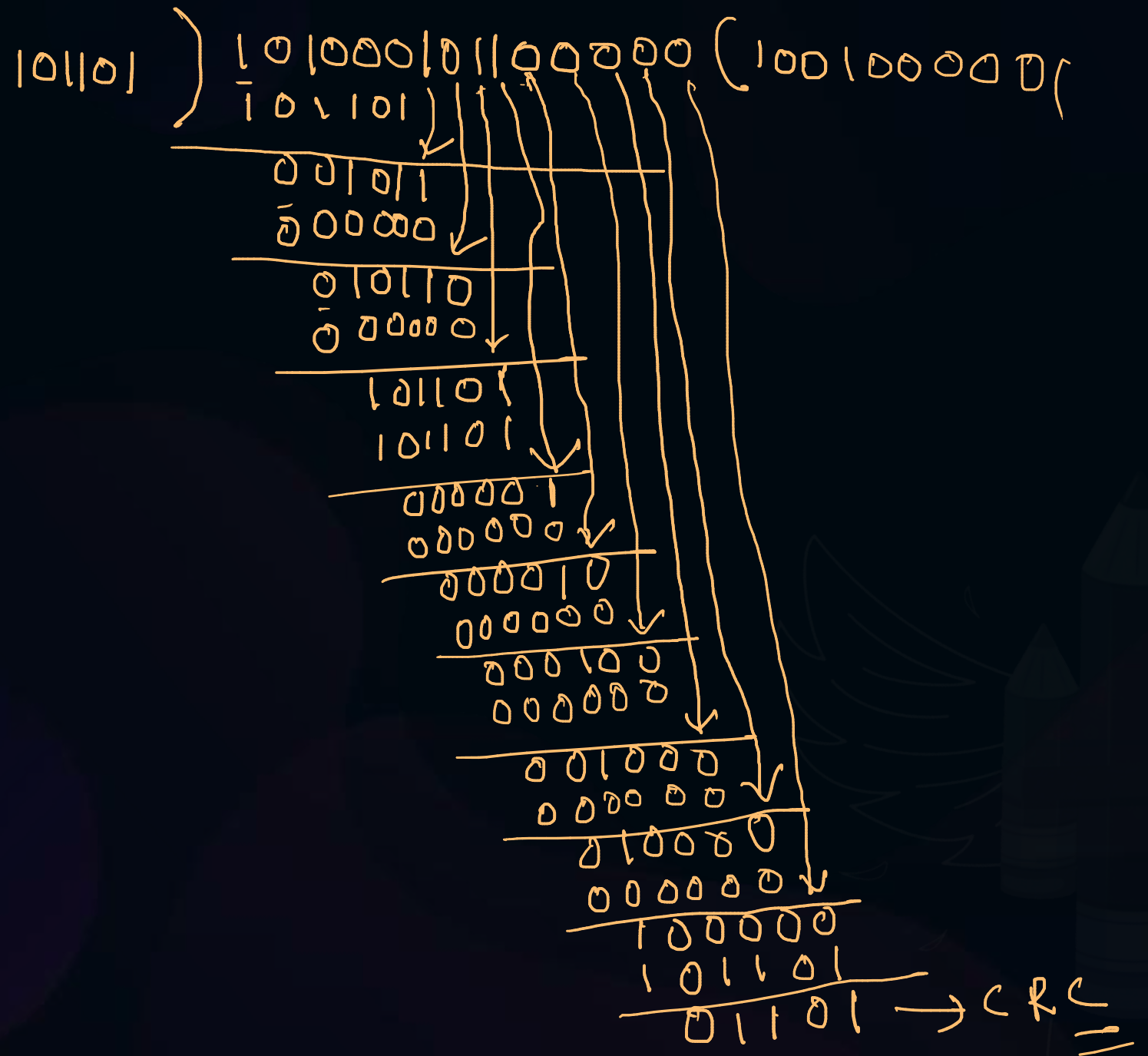
A 01110

B 01011

C 10110

D 01101

$x^5 + x^3 + x^2 + 1$ Polynomial → 101101 → Divisor.


CRC

[MCQ]

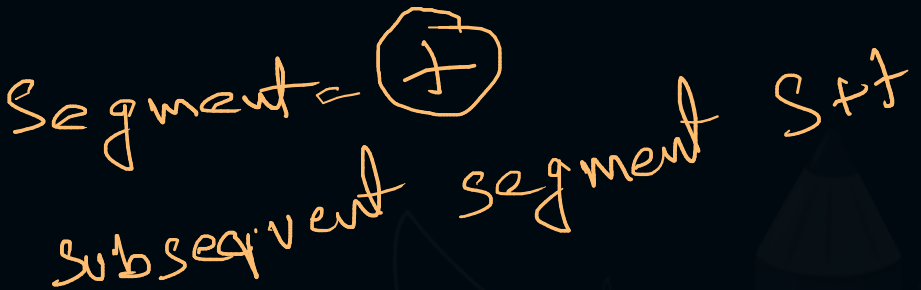
#Q. Consider a machine M1 that wants to send a huge file to machine M2 over a TCP connection. Suppose the sequence number for a segment in this connection is S, then, the subsequent segment's sequence number will necessarily be S + 1.

A. False

B. Depends on implementation

C. True

D. None



[MCQ]

#Q. Suppose there is exactly one packet switch between a sending host and a receiving host. The transmission rates between the sending host and the switch and between the switch and the receiving host are R1 and R2, respectively. Assuming that the switch uses store-and-forward packet switching, what is the total end-to-end delay in sending a packet of length L?
(Ignore queuing, propagation delay, and processing delay.)

A 2L/(R1 + R2).

B L/R1 + L/R2.

C 2L/R1* R2.

D L * R1 + L * R2.

Sender transmission rate $= R_1$

to transmit L length data packet by the sender $= \frac{L}{R_1}$

Switch transmission rate $= R_2$

to transmit L length data packet by the switch

$$= \frac{L}{R_2}$$

total end to end delay $= \frac{L}{R_1} + \frac{L}{R_2}$

THANK - YOU

